

Heat Transfer Homework 3:

Name: Paige Cobrin

Date of submission:

2/24/26

Problem Set number: 3

Due date:

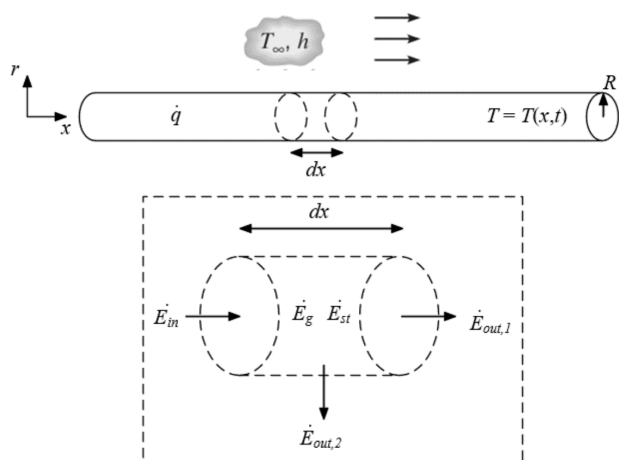
2/24/26

Acknowledgement: worked with
JT Thomforde

1. given: radius R
 heat rate $\dot{q}$
 environment: T_∞, h
 k, ρ, c_p

assumptions: uniform volumetric heat rate
 neglect thermal radiation
 radius of rod is small, temp along radial direction is uniform
 k, ρ, c_p constant

Schematic:



find: energy balance equation:

a) energy balance: $\dot{E}_{in} + \dot{E}_g - \dot{E}_{out,1} - \dot{E}_{out,2} = \dot{E}_{st}$

Annotations:
 $\dot{q}_x = -k A_c \frac{\partial T}{\partial x} \bigg|_x$
 $\dot{q} dV = \dot{q} A_c dx$
 $\dot{q}_x(x+dx) = \dot{q}_x + \frac{\partial \dot{q}_x}{\partial x} dx$
 $= -k_x \frac{\partial T}{\partial x} dy dz + \frac{\partial \dot{q}_x}{\partial x} dx$
 $\dot{q}_{conv} = h(dA_s)(T - T_\infty)$
 $\rho c_p dV \frac{\partial T}{\partial t}$

$$A_c = \pi R^2$$

$$dA_s = 2\pi R dx$$

$$-k A_c \frac{\partial T}{\partial x} + \dot{q} A_c dx - \left(-k_x \frac{\partial T}{\partial x} A_c + \frac{\partial \dot{q}_x}{\partial x} dx \right) - h dA_s (T - T_\infty) = \rho c_p A_c dx \frac{\partial T}{\partial t}$$

$$\dot{q} \pi R^2 dx - \frac{d}{dx} \left(-k \frac{\partial T}{\partial x} dx \right) \pi R^2 - h 2\pi R dx (T - T_\infty) = \rho c_p \pi R^2 \frac{\partial T}{\partial t} dx$$

$$\dot{q} + k \frac{\partial^2 T}{\partial x^2} - \frac{2h}{R} (T - T_\infty) = \rho c_p \frac{\partial T}{\partial t}$$

b) $\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) - \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$k \frac{\partial^2 T}{\partial x^2} + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

INITIAL CONDITIONS:

c) $T(0, t) = T_i$
 $T(x, 0) = T_i$

boundary conditions: $-k \frac{\partial T}{\partial x} \bigg|_{x=L} = h(T(x=L) - T_\infty)$

2. given: $L_A = 30 \text{ cm}$

$\dot{q}, k_A$

$L_B = 10 \text{ cm}$

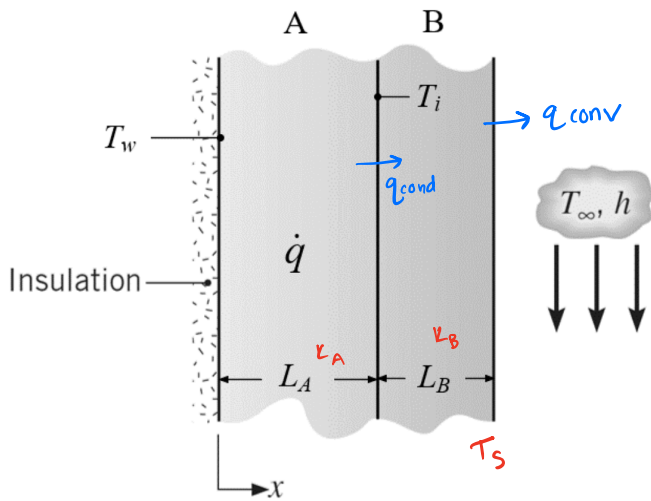
$k_B = 30 \text{ W/(m}\cdot\text{K)}$

$\dot{E}_{\text{gen}} = 0$

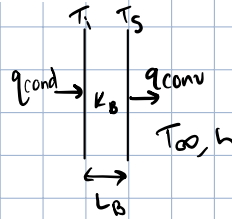
$T_{\infty} = 20^\circ\text{C}, h = 500 \text{ W/(m}^2\cdot\text{K)}$

$T_w = 115^\circ\text{C} \quad T_i = 100^\circ\text{C}$

assumptions: 1D heat conduction



find: a) T_s



$$q''_{\text{cond}} = q''_{\text{conv}}$$

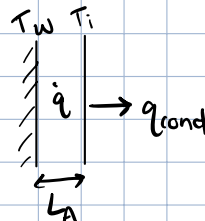
$$\frac{k_B (T_i - T_s)}{L_B} = h(T_s - T_{\infty})$$

$$k_B T_i - k_B T_s = L_B h T_s - L_B h T_{\infty}$$

$$T_s = \frac{k_B T_i + L_B h T_{\infty}}{(L_B h + k_B)} = \frac{(30)(373.15) + (0.1)(500)(293.15)}{(0.1)(500) + (30)} = 323.15 \text{ K}$$

$$T_s = 50^\circ\text{C}$$

b) $\dot{q}$ at steady state:



$$\dot{E}_{\text{gen}} = \dot{E}_{\text{cond}}$$

$$\dot{q} L_A A = A \frac{k_B}{L_B} (T_i - T_s)$$

$$\dot{q} = \frac{k_B (T_i - T_s)}{L_B L_A} = \frac{(30)(50)}{(0.1)(0.3)} = 50,000 \text{ W}$$

$$\dot{q} = 50 \text{ kW}$$

c) find k_A :

heat equation: $\frac{\partial}{\partial x} \left(k_x \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z \frac{\partial T}{\partial z} \right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$ ^{1D} ^{steady state}

$$k_A \frac{\partial^2 T}{\partial x^2} + \dot{q} = 0$$

$$\int \frac{\partial^2 T}{\partial x^2} = \int -\frac{\dot{q}}{k_A}$$

$$T = -\frac{\dot{q}}{2k_A} x^2 + ax + b$$

$$T = -\frac{\dot{q}}{2k_A} x^2 + 115$$

$$T(L_A, t) = T_i = -\frac{\dot{q}}{2k_A} (0.3)^2 + 115$$

$$100 = -\frac{50000}{2k_A} (0.3)^2 + 115$$

$$k_A = 150 \text{ W/(m}\cdot\text{K)}$$

BC:

$$\left. \frac{dT}{dx} \right|_{x=0} = 0$$



$$\frac{\partial T}{\partial x} = -\frac{\dot{q}}{k_A} x + a$$

$$a = 0$$

$$T(0, t) = T_w$$

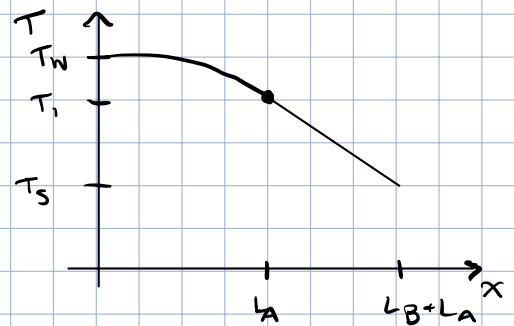
$$b = 115^\circ\text{C}$$

d)

$$0 < x < L_A: T = -\frac{\dot{q}}{2k_A} x^2 + 115$$

$$= -166.67 x^2 + 115$$

$$L_A < x < L_B + L_A: T = \frac{(T_S - T_i)}{L_B} (x - L_A) + T_i$$



3. given: $k_b = 25 \text{ W/(m}\cdot\text{K)}$
 $k_{TBC} = 1 \text{ W/(m}\cdot\text{K)}$

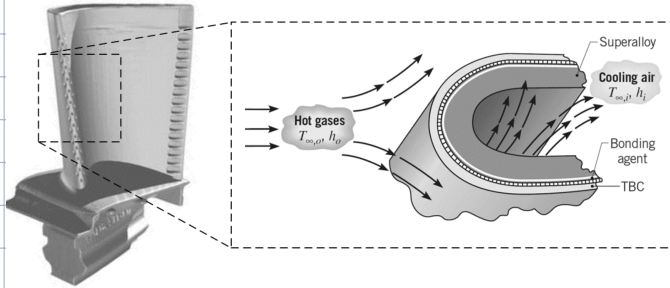
$R'' = 10^{-4} \text{ m}^2\cdot\text{K/W}$
 $T_{\infty,o} = 1616 \text{ K}$
 $T_{\infty,i} = 400 \text{ K}$

$h_o = 1000 \text{ W/(m}^2\cdot\text{K)}$
 $h_i = 500 \text{ W/(m}^2\cdot\text{K)}$

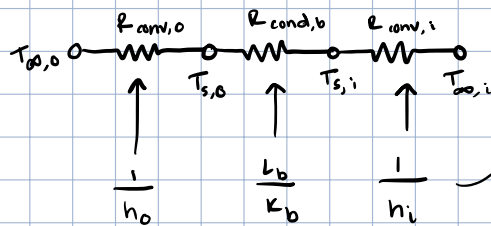
$L_b = 5 \text{ mm}$
 $T_{max} = 1200 \text{ K}$

assume: steady state turbine,
 neglect thermal radiation
 turbine blade ~ planewall
 $\rightarrow$ 2D $\dot{q}_{cond}$ through TBC and superwall

schematic:



a) with no TBC applied, find thermal resistance network, find ext. surface temp of superalloy



$$R_{tot} = R_{conv,o} + R_{cond,b} + R_{conv,i} = \frac{1}{1000} + \frac{0.005}{25} + \frac{1}{500} = 0.0032 \text{ m}^2\text{K/W}$$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R_{tot}} = \frac{1616 - 400}{0.0032} = 3.8 \times 10^5 \text{ W/m}^2$$

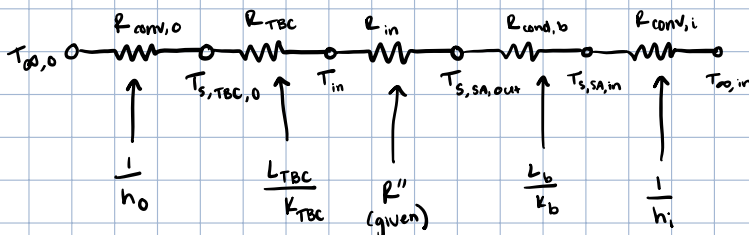
across outer surface: $q'' = \frac{T_{\infty,o} - T_{s,o}}{R_{conv,o}} = \frac{(1616 - T_{s,o})}{0.001} = 3.8 \times 10^5 \text{ W/m}^2$

$T_{s,o} = 1236 \text{ K} > 1200 \text{ K} = T_{max}$

$T_{s,o} = 1236 \text{ K}$

$\rightarrow$ NOT SAFE!

b) $L_{TBC} = 0.5 \text{ mm}$ find $T_{s,sa}$



$$R_{tot} = \frac{1}{h_o} + \frac{L_{TBC}}{k_{TBC}} + R'' + \frac{L_b}{k_b} + \frac{1}{h_i}$$

$$R_{tot} = \frac{1}{1000} + \frac{0.0005}{1} + 0.0004 + \frac{0.005}{25} + \frac{1}{500}$$

$R_{tot} = 0.0038 \text{ m}^2\text{K/W}$

$$q'' = \frac{(T_{\infty,o} - T_{s,TBC,o})}{R_{conv,o}}$$

$$q'' = \frac{\Delta T}{R_{tot}} = \frac{T_{\infty,o} - T_{\infty,i}}{R_{tot}} = \frac{1216 \text{ K}}{0.0038}$$

$3.2 \times 10^5 \frac{\text{W}}{\text{m}^2} \left(\frac{1616 - T_{s,TBC,o}}{0.001} \right) \rightarrow T_{s,TBC,o} = 1296 \text{ K}$

$q'' = 3.2 \times 10^5 \text{ W/m}^2$

$$q'' = \frac{(T_{s,TBC,o} - T_{in})}{R_{TBC}} = \frac{(1296 - T_{in})}{0.0005} = 3.2 \times 10^5 \rightarrow T_{in} = 1136 \text{ K}$$

$$q'' = \frac{(T_{in} - T_{s,SA,out})}{R''} = \frac{(1136 - T_{s,SA,out})}{0.0001} = 3.2 \times 10^5 \rightarrow T_{s,SA,out} = 1104 \text{ K}$$

$$T_{s,SA,out} = 1104 \text{ K} < 1200 \text{ K} = T_{max}$$

↳ SAFE!